

The Borrowable Standard

Why a corpus that must be authoritative in its own language cannot normalise toward someone else's — and what that made of one man's work in Cambodia between 1929 and 1969.

Authors: Thon Ly · Miss Aquarius (*Miss Aquarius is an AI co-author, disclosed; the inventor of record is Thon Ly*)

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Note on category. This paper fits none of the corpus's four categories cleanly. It is a history-of-infrastructure argument in support of an alignment corpus, filed under `alignment` because that is what it serves. It is **not** an essay, and putting it there would have placed a scholarly claim in the founder's personal voice. **The comparison cases in §8 have not been checked by the authors:** they are named as *the class this claim is testable against*, not as established support. The paper's own §4 exists because an unchecked assumption was checked and failed.

Preamble

A commission ordained in Phnom Penh in December 1929 translated the Pāli canon into Khmer, translating from 1930; the last of a hundred and ten volumes was printed at the end of 1968, and the work was celebrated in the capital on the first two days of April 1969. The man who had led it died that September. The man who succeeded him was executed in April 1975, on the second day of the Khmer Rouge's occupation of the city, at a pagoda in the district where he had been born.

This paper is not about that. It is about a smaller thing the same man did, on which, this paper argues, making the translation machine-readable as a witness depends: **he led the standardisation of how Khmer is written.**

The argument below began as a stronger and simpler claim — that a canonical digital corpus requires a stabilised orthography — and that claim is false. §4 shows why, using the counterexamples that refute it. What survives is narrower, and it is the reason the Cambodian case is not an instance of a general pattern but an exception to the way the general pattern is usually solved.

Prior-Art and Non-Assertion Statement

This document is dedicated to the public domain under CC0 1.0 Universal, and is published so that it stands as prior art. **Unlike most papers in this corpus, it specifies no mechanism** — the operating rules in §7.6 restate existing textual-scholarship practice for AI corpora — and what it discloses is the numbered list under *Findings disclosed*. No patent has been or will be sought on anything described in this paper by HeartBank®, Factory 333™, THonly™, Silicon WatSM, or any entity under their common control. **The authors and those entities commit not to assert any patent right against any party practising anything disclosed here.** The commitment is stated rather than implied, and is not conditioned on reciprocity, attribution, or field of use.

The literature this paper stands on, and does not claim to originate:

- **Language planning.** Haugen's four-stage model — *selection · codification · elaboration · acceptance* (1966; the 1983 revision reads *implementation*) — is the frame within which everything here sits. What Chuon Nath performed is **codification** in Haugen's sense, and this paper claims no new theory of codification; the two-layer account of §7 is an observation about what codification does not settle.
- **Kloss's (1969) distinction between *corpus planning* (the form of a language) and *status planning* (its social position).** A terminological warning the reader is owed immediately: **"corpus" in the language-planning sense and "corpus" in the computational sense are different words.** This paper argues that *corpus planning* in Kloss's sense is a precondition for a *corpus* in the NLP sense, and the pun is unfortunate but the connection is real.
- **Low-resource NLP and dialect processing** — the Swiss German and dialectal Arabic literature that §4 uses against this paper's own first formulation.
- **Script encoding.** The Unicode Khmer block (U+1780–U+17FF), and the general problem of legacy 8-bit font encodings preceding it.

No claim of priority is made over any of the above. The contribution offered is in §6 and §7: a distinction between corpora that may normalise toward a borrowed standard and corpora that may not, and a two-layer account of what standardisation does and does not buy.

Abstract

This study concerns orthographic standardisation, text normalisation and legacy character encoding as preconditions for machine-readable text corpora that must stand as witnesses in a critical apparatus, with Khmer and the Pāli canon (Tipiṭaka) as its case.

It is widely assumed, and rarely stated, that digitising a language is a tooling problem — better optical character recognition, better fonts, better models. This paper argues that for one class of corpus it is not, and that the constraint is upstream of any tool.

We begin from a claim that turns out to be **false**: that a machine-readable corpus requires the language to have a standardised orthography. Swiss German and the Arabic dialects refute it. Both lack standardised

spelling; both have substantial corpora, speech systems, machine translation and dedicated benchmarks. Corpora can be and are built for languages whose spelling is unsettled.

What the refutation reveals is the mechanism by which those corpora are possible: they normalise toward a standard the variety itself does not have. Swiss German normalises to Standard German. Dialectal Arabic normalises to Modern Standard Arabic, or to a convention created for the purpose. The corpus is *about* the variety and *encoded* in a target borrowed from elsewhere.

That move is available to most languages and unavailable to a specific and important minority: **a corpus that must be authoritative in its own language cannot normalise toward another.** A Khmer canonical text rendered in Thai orthography is not a Khmer witness to anything; it is a Thai rendering, and the distinction is the entire content of a critical apparatus. For such corpora, **native standardisation is a genuine precondition, because the borrowing that substitutes for it is disqualified by what the corpus is for.**

We then examine the Cambodian case, which is unusual in that the precondition has a name, a date and a documented programme. Between 1926 and 1967 Chuon Nath led the standardisation of Khmer orthography, compiled the dictionary that fixed its lexicon, and led the translation of the canon into it. **But the case also contains its own counterexample, and this is the paper's second contribution: standardisation is necessary and not sufficient.** Khmer's orthographic layer was codified in its dictionary by 1967; its **encoding** layer was not, and a Khmer volume this project holds — from a translation of the canon with its commentaries, first printed in 2009 — carries **five mutually incompatible legacy font encodings in a single book**, one of them apparently custom and undocumented. Nearly sixty years separate the codification of the first layer from the ongoing repair of the second.

We state the claim in falsifiable form, name the comparison class against which it should be tested, and concede three limits — including that standardisation was contested, that its politics are not innocent, and that this paper is not an argument for it.

Keywords: orthographic standardisation (standardization); text normalisation (normalization); spelling normalisation; collation; critical apparatus; textual witness; legacy font encoding; Unicode Khmer; character encoding; low-resource language corpora; dialect corpora; Swiss German; dialectal Arabic; CODA; language planning; corpus planning; codification; Pāli canon; Tipiṭaka; Khmer; citation-grounded AI; provenance.

Terms

The paper's own phrases, mapped to the standard terms a reader in textual scholarship, language planning or natural-language processing would use.

Term in this paper	Standard term
corpus <i>as witness</i> ; corpus "authoritative in its own language"	a transcription that stands as a witness in a critical edition (documentary or diplomatic transcription)
corpus <i>about</i> a language	a general-purpose text corpus for training or evaluation
borrowed target; borrowable standard	normalisation target in a related standard language (e.g. Standard German, Modern Standard Arabic)
a convention built for the purpose	conventional orthography for computational use (e.g. CODA for dialectal Arabic)
rendering	normalised, transliterated or re-scripted text standing in place of the source form
replacement normalisation	destructive (irreversible) text normalisation
normalisation kept as a second column	a parallel normalisation layer that retains the original form (TEI <code><choice></code> with <code><orig></code> and <code><reg></code>)
lexical layer · orthographic layer · encoding layer	lexicon (dictionary of record) · orthography (spelling standard) · character encoding
per-font transcoding table	legacy-font-to-Unicode conversion table
the missing column; <code>ed="K"</code>	a page-break concordance column for a print edition (TEI <code><pb ed="..." /></code>)
siglum (<code>sī.</code> , <code>syā.</code> , <code>kaṃ.</code> , ...)	witness siglum in a critical apparatus
codification	codification in Haugen's model of language planning; corpus planning in Kloss's terms
<i>saṅgāyana</i>	communal recitation council (Pāli); the Sixth Council of 1954–56 produced the Chaṭṭha Saṅgāyana edition
Thommakay · Thommaboran	the reformist ("new dhamma") and traditionalist ("ancient dhamma") currents within Cambodia's Mahānikāya order
the fourth failure mode (§8.5)	source substitution in citation-grounded generation: a citation that resolves to a rendering, not to the source

Findings disclosed

The paper discloses no mechanism. Its findings are listed here, each with what would break it; the sections named carry the argument and the evidence. Findings 5 to 7 come from material the authors hold and are scoped to it (§9.5, §9.5.1).

1. **Text corpora for varieties without a standardised orthography are built by normalising toward a standard borrowed from a related language, or toward a convention built for the purpose** — Swiss German toward Standard German (the SwissText 2021 shared task; SDS-200; Aepli et al. 2023), dialectal Arabic toward Modern Standard Arabic or toward CODA (Habash, Diab & Rambow 2012) (§4–§5).

Breaks if: a substantial corpus for an unstandardised variety is found that uses no normalisation target of either kind, in its annotation or in its evaluation.

2. **For a corpus that must be authoritative in its own language — a witness to a text, not a corpus about a language — normalising toward a borrowed target disqualifies the corpus, so native orthographic standardisation is a precondition for it (§2, §2.1, §6).** Stated in its strong form; one case cannot separate it from the weaker reading that native standardisation makes such corpora much easier to build (§9.4.1). *Breaks if:* the test of §8 is met — a language with no native orthographic standard and no adequate borrowable neighbour that nonetheless supports a corpus treated as authoritative in that language.
3. **Normalisation that replaces source forms is the inverse of collation: replacing first and collating second yields an apparatus of normalisation artefacts rather than of traditions, while a normalisation kept as a parallel layer beside the retained source form does not (§6.1).** *Breaks if:* a collation run over replacement-normalised witnesses recovers the inter-witness differences that the replacement removed, without access to the source forms.
4. **Standardising a language's written form and standardising its digital representation are separate acts, separable by generations, and the first is necessary but not sufficient for the second:** a legacy-font-to-Unicode conversion is well-defined only against a target orthography, and Khmer's orthography was codified in its dictionary by 1967 while its encoding was not (§7.1–§7.3). The lexicon is a third layer, settled or borrowed independently (§7.1.1). *Breaks if:* a legacy encoding is shown to map determinately to Unicode for a script whose orthography is unsettled, or the Khmer encoding layer is shown to have been settled together with its orthography.
5. **One Khmer volume held by this project — from a translation of the canon with its commentaries, first printed in 2009 — embeds five legacy Khmer font encodings, one of them (Bidokk1) apparently cut for the series and undocumented (§7.2).** A fact about that volume, not a survey (§9.5). *Breaks if:* the volume's embedded-font list differs, the five names resolve to fewer distinct glyph-to-codepoint mappings, or a published mapping for Bidokk1 is found.
6. **In the Chaṭṭha Saṅgāyana Tipiṭaka's XML release, the Cambodian edition has no page-concordance column — no <pb ed="K"> in any of its 217 files — while the variant apparatus does cite the Cambodian siglum kaṃ. , in about 8 per cent of its notes (1,820 of 22,343 in the 2026-09-05 count; 1,859 of 22,345 in the 2026-10-01 re-count) (§7.5).** *Breaks if:* an ed="K" page marker is found in the release; a decoded count of the notes citing kaṃ. differs materially; or ka. (7,266 notes), which tipitaka.org's help page also lists as Cambodian, proves to denote a Cambodian source, in which case the Cambodian witness is not sparse.
7. **Of two scan sets of the same Khmer edition, the cleaner one had lost the per-volume front matter that carried both the scan's provenance and its distribution dedication: cleaning pipelines remove what is not content, and provenance is not content (§7.7).** *Breaks if:* the removed pages prove present in the cleaner set, or to carry neither provenance nor permission.
8. **A citation-grounded system has a failure mode that model-level and retrieval-level checks cannot detect: the citation resolves, the source supports the claim and is correct, and the encoded text is a rendering rather than the source (§8.5).** *Breaks if:* a citation-accuracy evaluation detects the substitution of a rendering for a witness without access to the encoding's provenance.

9. In the Khmer case the translation, the dictionary and the orthography ran as mutually dependent programmes, not a one-way sequence — an orthography committee from 1926, a dictionary first published in 1938 and made definitive in 1967, a translation from 1930 to 1968 (§7.1). The further reading of the Thommakay reform's two commitments as one is an interpretation, not a documented motive. *Breaks if*: the dated record shows any of the three finished before the others began.

1 · Why a corpus project is asking a question about spelling

This paper is a by-product of building something. The institution publishing it is assembling a machine-readable Khmer witness to the Pāli canon — the Cambodian recension, entered into a critical apparatus that collates Sinhalese, Siamese and PTS readings against its Burmese base text, in which the Cambodian siglum is sparse and the Cambodian edition has no page concordance at all (§7.5).

The work is not textual scholarship for its own sake. It exists because an alignment programme needs a **value corpus whose every claim resolves to a citation** — work, edition, locus, and the lineage by which that edition is attested. A system that cannot name where a judgment comes from is a system whose judgments cannot be checked, and checkability survives its author where trust in the author does not.

That requirement is what made spelling a first-class problem rather than an implementation detail. A **citation is a promise that a reader can go and look**. If the text a reader would find is spelled differently from the text the system indexed — not *wrongly*, but under a different convention — the promise fails silently, and it fails in a way no amount of retrieval engineering repairs.

1.1 · The shape of the problem, before any of it was about spelling

The requirement arrived from the other end. An alignment programme wanted a values model whose every judgment resolves to a citation — and that requirement, pursued honestly, decomposes:

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"every judgment cites its source"
├── the source must be IDENTIFIED      → work, edition, locus
├── the identification must be CHECKABLE → a reader can go and look
└── what the reader finds must be      → and here the requirement
    the same text the system indexed    stops being about software
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The first two are engineering. **The third is not**, and it is where a project discovers that its corpus has a history it did not choose. A citation into a text encoded under a different orthographic convention from the one a reader will consult is a citation that resolves to a *different object* — and no part of the system can detect that, because every component is behaving correctly.

This paper is therefore a report from the point where an infrastructure question surfaced inside a values question — the point at which, by the time anyone notices, the decision has usually been made by someone else and belongs to someone else's field.

The bounded ambition, stated so the reader can hold us to it: this paper does not argue that standardisation is good, that Cambodia's case is representative, or that anything follows for languages in

general. It identifies a **precondition for one class of corpus**, exhibits a case where that precondition has a documented author, and says what would falsify the claim.

2 · The claim, stated precisely

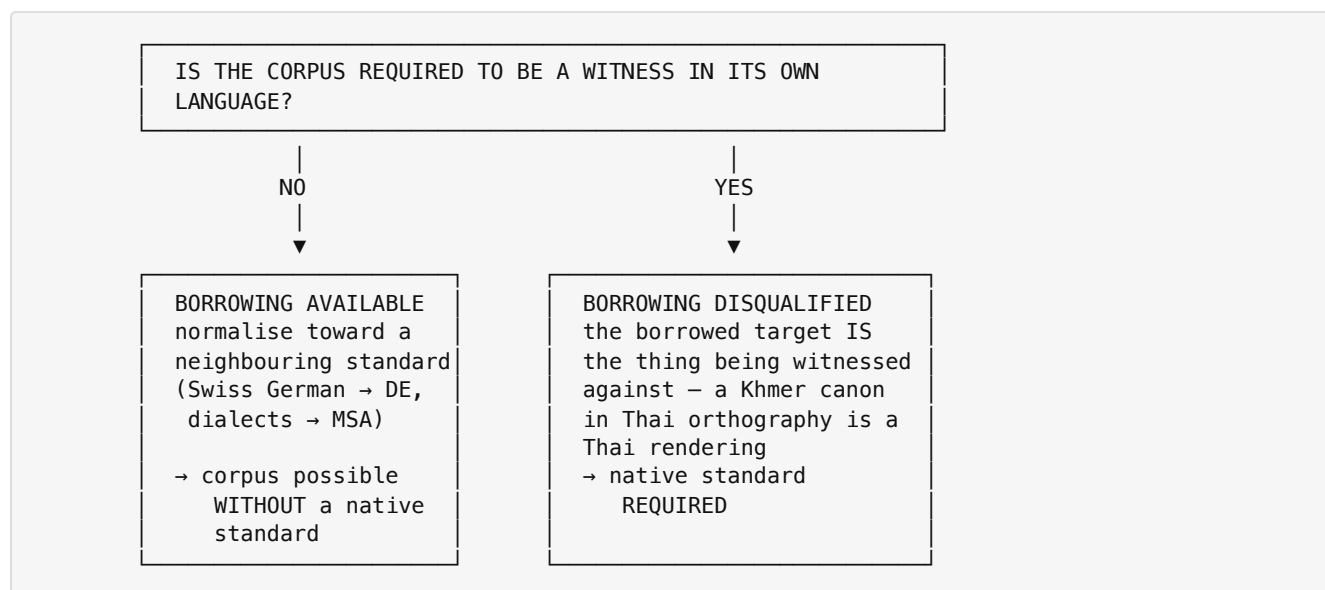
For a corpus that must be authoritative in its own language, native orthographic standardisation is a precondition — because the substitute that makes other corpora possible, normalising toward a borrowed target, is disqualified by what such a corpus is for.

Three terms are doing work.

"**Authoritative in its own language**" means the corpus is not merely *about* the language but is required to *be* a witness in it — such that a reader, a scholar or a court could treat the encoded text as standing for what the source says. Canonical religious texts, statutes, treaties and critical editions are of this kind. A sentiment-analysis dataset is not.

"**Native**" distinguishes standardisation of the language itself from adoption of a neighbour's: codified for this language by its own users, whatever institution convened them. An **adequate** borrowable target is a neighbour's standard whose forms the variety's readers accept as a rendering of their own language; and a corpus is **authoritative** when a scholar or a court would treat the encoded text as standing for the source.

"**Precondition**" is stronger than *helpful* and weaker than *sufficient*. §7 is devoted to the gap between those two. *The claim is stated in its strong form because that is the form worth attacking (§3); §9.4.1 records what the evidence can and cannot separate.*



2.1 · What "authoritative" is doing, since the whole claim turns on it

The distinction is between a corpus that is **about** a body of text and a corpus that **is** an instance of it, such that a third party may reason from the encoding as they would from the source.

	corpus about	corpus as witness
Question it answers	<i>what do texts of this kind look like?</i>	<i>what does THIS text say, here?</i>
Tolerates normalisation	yes — often requires it	no
Tolerates silent regularisation	yes	no
A wrong character is	noise, absorbed by scale	a false reading, propagated as fact
Recoverable from the encoding	the phenomenon	the source itself

Sentiment corpora, language-identification sets and most training data are of the first kind, and this paper says nothing about them. Statutes, treaties, contracts of record, critical editions and canonical religious texts are of the second. **The population is small, old, and disproportionately consequential** — and it is exactly the population an AI system needs when it is asked to say what a tradition, a law or a contract actually holds.

One clarification the reader is owed: "authoritative" here is a claim about the ENCODING's relationship to a source, not about the source's own authority. Whether the Cambodian recension is *correct* is not this paper's business; whether an encoding of it is *that recension* is.

3 · The claim we started with, and why it is worth showing

The first formulation was: **a machine-readable canon requires a stabilised orthography.** It is clean, it matched the Cambodian evidence, and it is wrong.

We report it because the manner of its failure is the paper's method. The claim was tested adversarially before drafting — not by looking for cases that supported it, but by looking for the case that would kill it. The case exists, it was found in an afternoon, and the thesis that replaced it is narrower, more useful and falsifiable in a way the original was not.

A claim that has not been hunted is a claim whose supporting cases were selected.

4 · The counterexample: corpora do get built without standardised spelling

Swiss German has no standardised orthography. Writers spell dialectally and inconsistently; there is no official codification of the written form. It nonetheless has a substantial resource ecosystem — dialect corpora assembled from text messages and from archival speech, corpora with annotated normalisation and part-of-speech tagging, parallel multidialectal speech corpora, and speech-to-text corpora at scale.

The Arabic dialects are the same case at greater scale. Egyptian, Levantine and Gulf varieties lack standard orthographies and are predominantly spoken, and there is a long research literature on tagging, translating and modelling them regardless.

There is even a dedicated benchmark literature on evaluating machine translation **for dialects without standard orthography** — which is a field admitting, in its own title, that the condition is normal and

workable.

4.1 • What the literature actually reports, since the detail matters for §5

Three features recur across that work, and all three are consistent with the reading offered in §5 rather than with orthographic standardisation being irrelevant.

Normalisation appears as an explicit annotation layer, not as a preprocessing convenience. Dialect corpora are built with a normalised form recorded *alongside* the dialectal transcription, which is the practice of a field that knows the two are different objects and needs both.

Spelling variation is treated as a first-class research problem rather than as noise to be absorbed — dedicated work on dialect variation dictionaries, on mining variant spellings from social media, and on evaluating translation quality where no reference orthography exists.

And evaluation itself becomes contested. There is benchmark work specifically on *how to evaluate machine translation for dialects without standard orthography*, which is a field acknowledging that the absence of a standard breaks the measurement instrument, not merely the training data.

None of that is a field working comfortably without standardisation. It is a field managing its absence, at cost, by leaning on a standard next door.

So the strong claim is refuted, and this paper does not attempt to rescue it. Corpora can be built for languages whose spelling is unsettled. Anyone asserting otherwise is contradicted by a decade of published work.

5 • What those projects actually do — and it is not "cope with variation"

The interesting question is not *whether* they succeed but *how*, and the answer is consistent across both families.

5.1 • They normalise toward a standard the variety does not have

Swiss German resources largely normalise to **Standard German** (the SwissText 2021 shared task; SDS-200; Aepli et al. 2023). Corpora carry an explicit normalisation layer alongside the dialectal transcription; speech corpora are built as *Swiss German speech* → *Standard German text*. The dialect is the **source**; the standard is the **target**.

Dialectal Arabic normalises to **Modern Standard Arabic**, which occupies the same structural position — a written standard, universally taught, not natively spoken in the relevant sense, and available to normalise toward.

DIALECT (no standard)	→	NORMALISATION layer	→	ENCODED FORM (a standard the dialect does not itself have)
Swiss German	→	map to Standard German	→	
Egyptian Arabic	→	map to MSA / convention	→	
Khmer canon	→	???	→	nothing to map to

The standard is not dispensed with. It is borrowed. The corpus is possible because a codified target exists nearby, not because codification turned out to be unnecessary.

5.2 • And where nothing adequate exists to borrow, they build one

The dialectal-Arabic research community, finding MSA an imperfect target for dialectal forms, developed **CODA**, a conventional orthography for dialectal Arabic (Habash, Diab & Rambow 2012) — created deliberately, its authors state, because dialectal Arabic has no standard orthography: a standard built where none could be borrowed.

It is the strongest support this paper has found for its thesis, and it comes from the community that refutes the paper's first version. Faced with no adequate borrowable target, the response was **not to abandon standardisation but to perform it** — decades later, by different people, for the reason this paper turns on: no adequate target existed to borrow.

6 • Why a canonical text cannot borrow

Borrowing has a cost, and for most corpora it is acceptable.

Normalising Swiss German to Standard German discards the dialect. The corpus becomes a record *about* a variety, encoded in another. For machine translation, speech recognition and most downstream tasks this is not merely tolerable but desirable: the standard form is what the application needs.

For a corpus required to be a witness, the same move destroys the thing being built.

Consider the concrete case. The Cambodian recension of the Pāli canon differs from the Burmese in readings — that difference is precisely what a critical apparatus records, and precisely what a Khmer witness would contribute to an apparatus in which it is now sparse (§7.5). Now normalise it toward a neighbouring standard. **What has been encoded is no longer a Khmer witness.** It is a rendering, and a rendering inherits the lineage of what it was rendered *into*, not what it was rendered *from*.

6.1 • Normalisation and collation are inverse operations

The mechanism deserves stating exactly, because it explains why the loss is total rather than partial.

A critical apparatus records differences between witnesses. That is its entire function: at locus *Z*, witness *K* reads *X* where witness *M* reads *Y*. The value of a new witness is precisely the differences it contributes.

Normalisation is the operation that removes differences. It maps variant forms onto a canonical form so that downstream processing sees one thing where the source had several. That is why it works so well for the corpora of §4 — variation is noise there, and removing it is the point.

COLLATION

preserves difference
output: an apparatus
value $\propto$ variation retained

NORMALISATION

removes difference
output: a canonical form
value $\propto$ variation eliminated

Replacing source forms before collating destroys collation's input.

So normalisation that *replaces* the source form with a borrowed standard, and then collates, is not lossy — it is empty. A normalisation kept as a second column beside the retained source form — the reading-and-witness structure the TEI critical-apparatus module provides — is collation's *input*, not its destroyer. The disqualifying move is the replacement, and it is the move the tractable pipelines make by default. The differences that would have constituted the witness were removed by the step that made the corpus tractable. **The apparatus that results records the differences between normalisation artefacts, not between traditions.**

And this is why the failure is invisible. A collation run on normalised text produces output. It has entries. It looks like an apparatus. Nothing in the pipeline reports that the input was pre-flattened — because normalisation is not an error, it is a successful operation that was simply inappropriate here.

This is not a subtle point and it has an ugly failure mode: the rendering looks correct. It is readable, it is in a recognisable script, and it corresponds passage-for-passage to the source. It simply attests nothing. A project that mistakes one for the other produces a corpus that is confident and empty — and if that corpus is then used to ground a system's claims about what a tradition says, the error propagates into every judgment the system emits.

So the general pattern — borrow a target — is available to most corpora and closed to this one. Where it is closed, the precondition it substituted for returns.

7 · The Cambodian case, in two layers

The case is unusual not because standardisation happened — it happens often — but because it happened **recently, in one documented programme, with a named leader**, and because the corpus it enabled is still being built by people who can read the primary sources. Most standardisations are reconstructed by historians. This one has living successors.

And it contains its own counterexample, which is the more useful half.

7.1 · Layer one — the orthographic layer, codified

Chuon Nath (1883–1969) was Sangharāja of the Mahānikāya and led the reformist current within it, the **Thommakay**, against the traditionalist **Thommaboran**. The reform had two commitments that matter here: **direct study of the Pāli source** rather than doctrine received through tradition, and **adoption of modern tooling**, the printing press above all.

Those two commitments are usually listed as separate modernising sympathies. We read them as one commitment — an interpretation, not a documented motive. *Go and read the source yourself* is an

instruction that cannot be given to a community whose sources exist as individually hand-copied palm-leaf manuscripts — not because people cannot read, but because **there is no shared object to point at**. Two readers consulting two manuscripts are not consulting the same text, and neither can correct the other by reference to it.

The printing press is what turns go and read it from an exhortation into an instruction. Mass identical reproduction creates the shared object that makes source-primacy actionable, and it is why a reform grounded in direct study of the canon needed a press rather than merely welcoming one.

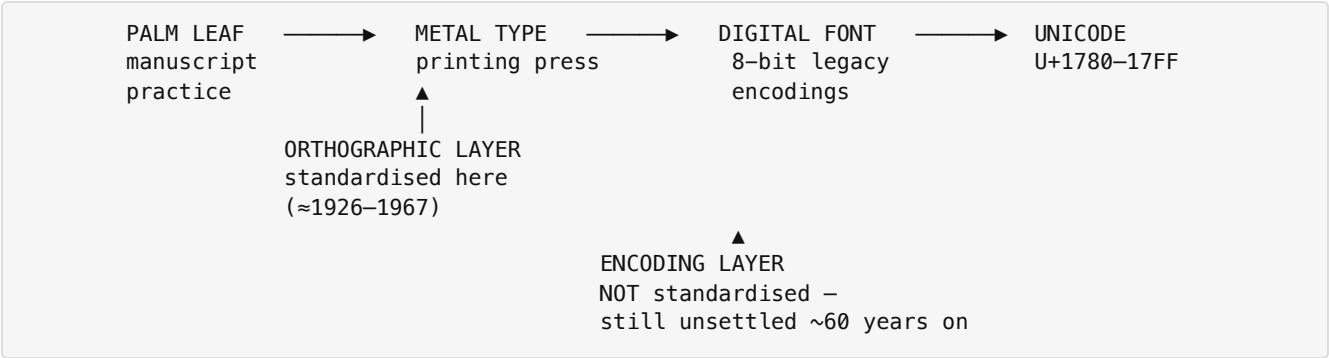
That logic is worth pausing on, because it recurs. A commitment to checkability generates a demand for whatever technology makes the checked thing identical across checkers — and each time that technology arrives, it brings a new standardisation problem with it. The press required a settled orthography. Digital encoding required a settled character set. Neither requirement was visible until the technology made it binding.

Within that programme he did three things which are usually described separately and are better understood as one:

The translation	the Tipiṭaka Commission, 1929–1968, 110 volumes
The dictionary	first published 1938; definitive edition 1967
The orthography	standardisation of Khmer writing — layout, symbols, punctuation

These are not three achievements. They are one achievement with three faces: you cannot translate a canon into a vernacular without deciding what its words mean, and you cannot print the result without deciding how they are written. **The three programmes ran together and each forced revisions in the others** — the orthographic decisions began in a committee founded by royal ordinance in July 1926, which Chuon Nath joined (Sasagawa 2015), before the Commission was ordained in 1929; the dictionary was under way before its 1938 first edition; and its definitive edition (1967) postdates most of the translation it stabilises. The dependency is mutual, not one-way.

That is worth stating carefully, because it converts an observation about a remarkable individual into something a person can act on: **convergent contributions of this kind arise from attempting one thing and refusing to leave its preconditions unbuilt.**



7.1.1 · There is a third layer, and it is the one usually left implicit

This paper has argued two layers. The Cambodian case exhibits three, and the middle one is the easiest to overlook because it does not look like infrastructure.

LEXICAL LAYER	what the words MEAN	the dictionary
ORTHOGRAPHIC LAYER	how they are WRITTEN	the standard
ENCODING LAYER	how they are STORED	Unicode / legacy

each can be settled or unsettled independently of the others

A translation into a vernacular is not well-defined until the vernacular's terms are. Rendering a technical Pāli vocabulary — *saṅkhāra*, *upādāna*, *paṭicca-samuppāda* — into Khmer requires deciding what the Khmer terms will mean, and those decisions are only stable if they are recorded somewhere a later reader can consult. **The dictionary is that record**, and its 1967 edition postdates most of the translation it stabilises.

Which produces a fact this project depends on operationally, not just historically: the dictionary is the TRANSLATOR'S OWN LEXICON. It is not a general reference that happens to be Khmer. It is the semantic key to the translation choices in the very edition being transcribed, compiled by the person who led the making of them.

For a Pāli↔Khmer parallel corpus that is a rare instrument. Where a Khmer term renders a Pāli one, the *intended* gloss is available rather than inferred — and where legacy-font extraction yields a damaged near-word, a lexicon of the same register and period — 17,328 entries by our transcription's count of the 1967 edition; published counts differ by edition and by what is counted as an entry — is a far better disambiguator than a modern general dictionary.

The layers can be borrowed independently, and this is where the §5 pattern generalises. A project can borrow an encoding (adopt Unicode), borrow an orthography (normalise to a neighbour), or borrow a lexicon (gloss through a third language). **Each borrowing is disqualifying for a witness for the same reason, and the third is the least visible** — a corpus glossed through English or Thai carries those languages' semantic joints into a text that did not have them.

7.2 · Layer two — the encoding layer, unsettled for decades

Standardising how a language is *written* does not standardise how it is *stored*. Between the metal type and Unicode sits a period in which Khmer was set digitally using **legacy 8-bit fonts that mapped Khmer glyphs onto Latin codepoints** — a general phenomenon for non-Latin scripts before Unicode adoption, and an acute one for Khmer.

Volumes this project holds exhibit it directly. **A single volume of a Khmer translation of the canon with its commentaries, first printed in 2009 and held by this project for study rather than for transcription, embeds five distinct Khmer font encodings** — a Limon-family face, APSARA, ThoeunA1, TacteingA, and Bidokk1, the last apparently cut for this series and, so far as we can determine, undocumented. Extracting text yields Latin character sequences that are not Latin words:

```
extracted: brmtṚeCatika Gdækfa kfaBN'naéRtsrN³  
intended: the same title in Khmer script (Unicode block U+1780–U+17FF),  
romanised: Paramatthajotikā aṭṭhakathā ...
```

Recovery requires a per-font transcoding table, applied per text-run, and the transcoding is lossy in font-dependent ways at exactly the points where Khmer orthography is most structured — the subscript *coeng* clusters and the reordering vowels. A second body of material, born-digital and carrying a genuine Unicode text layer, fails differently: subscripts dropped, vowels detached from their bases, and **Private Use Area codepoints** where glyphs were never mapped at all.

These are two different repairs and must not be assumed to be one pipeline. The first is a font-transcoding problem; the second is a glyph-reordering and PUA-recovery problem.

7.3 · What the two layers together actually show

Standardisation is necessary and not sufficient, and the Cambodian case demonstrates both halves at once:

- **Necessary**, because the transcoding tables of §7.2 are only *well-defined* against a target orthography. A legacy encoding can be mapped to Unicode because there is a determinate correct answer to map to. **Without layer one, layer two has no destination**, and the repair is not merely hard but undefined.
- **Not sufficient**, because layer one was codified by 1967 — and, after a partial revision of 1972 had held schools and the press for decades, officially revived in 2009 (§9.2) — while layer two is still being paid for in 2026. Nearly **sixty years** separate the codification from that repair.

The general shape, stated for reuse: standardising a language's *form* and standardising its *digital representation* are distinct acts that can be separated by generations, and a project that inherits the first without the second inherits a well-defined problem rather than a solved one. *That is a better position than an ill-defined problem, and it is not the same as being finished.*

7.5 · What the missing column looks like, concretely

The abstraction becomes checkable at this point, and the numbers are worth stating because they show the shape of the absence.

A widely used digital edition of the Pāli canon is the **Chaṭṭha Saṅgāyana Tipiṭaka**, the recension settled at the Sixth Buddhist Council in Yangon, 1954–56, published digitally as TEI XML (the romanised release distributed by tipitaka.org, as held by this project on 2026-09-05; the same release is published at `VipassanaTech/tipitaka-xml`, directory `romn/`). Its files carry two independent pieces of apparatus machinery:

Machinery	What it records	Coverage
Page concordance (<code><pb ed="..." /></code>)	where a locus falls in each print edition	V VRI · M Burmese · T Siamese · P PTS
Variant apparatus (<code><note></code>)	where witnesses disagree, with sigla	sī. Sinhalese · syā. Siamese · pī. PTS · kaṃ. Cambodian · ka. (disputed; see below)

Across the full set of 217 files — root texts, commentaries, sub-commentaries and other Pāli works — the concordance carries page markers for four print editions (a fifth marker, 0, occurs only in fifteen later Pāli works — chronicles, poems and grammars), and the apparatus carries **22,343** recorded variant readings in the count made on 2026-09-05. The files are UTF-16 and must be decoded before they are counted: a byte-level `grep` for `<note>` finds none, and a count of matching lines undercounts, because one line can carry several notes. A re-count of decoded `<note>` elements on 2026-10-01, on the same files (byte-identical to `VipassanaTech/tipitaka-xml`, `romn/`, commit `05d5d3c`), finds 22,345; the small differences from the 2026-09-05 figures are differences of counting method, not of text.

The count for the Cambodian siglum kaṃ. is 1,820 — not zero (1,859 notes in the 2026-10-01 re-count; a bare search for `kaṃ.` also matches Pāli words ending in *-kaṃ*, so only notes are counted). An earlier revision of this paper stated *the count is zero* from a twenty-five-file sample searched with a tool that treated the files as binary; that sample also carried no `kaṃ.` note, and the zero read exactly like an absence. It was a probe failure, and it is withdrawn here rather than quietly fixed. What is absent is the Cambodian edition from the **page concordance** — no `<pb ed="K">` exists, so a Khmer locus cannot be cited by page — and the `kaṃ.` witness in the apparatus is sparse beside the Sinhalese and Siamese ones: about 8 per cent of notes in either count, against about 39 and 37 per cent in the re-count. **The missing column is therefore a concordance column and a fuller witness, not a first witness.**

One reading in that account is unsettled, and it bears on sparse. `tipitaka.org`'s help page lists `kaṃ.` and `ka.` together as Cambodian. In the files the two attest different readings in 499 notes, and one note names them as separate sets of books (*syā. kaṃ. ka. potthakesu*), so they do not denote a single witness. An earlier specification of this project read `ka.` as *kesuci*, "in some manuscripts", and no published key that says so has been found. If `ka.` (7,266 notes in the re-count) also denotes a Cambodian source, the Cambodian witness in the apparatus is not sparse and the paragraph above overstates the gap; the concordance finding does not depend on it.

This is not an oversight and should not be described as one. The Sixth Council was convened in Burma, working from Burmese manuscripts, with the Sinhalese, Siamese and PTS editions as its principal comparanda. **Cambodia's sparseness is the geography of 1954** — Cambodian elders took part in the Council, and a Cambodian witness was consulted, but far less often than the others, and the files do not say which Cambodian books the siglum denotes — fourteen years before the Khmer edition finished printing, and two decades before the Khmer Rouge took Phnom Penh.

7.5.1 · And the apparatus in which Khmer is sparse was itself produced by an act of the same kind

The Sixth Council was a **saṅgāyana** — a communal recitation, in which the canon is recited in assembly and discrepancies are resolved. The recension it produced, and the apparatus that records where it differs from the Sinhalese and Siamese lines, are the output of a deliberate collation event convened for exactly that purpose.

So the structure this project proposes to add a column to is not a neutral container. It is the artefact of one tradition's standardisation act, carrying its comparanda at the weights it gave them. That is not a criticism — an apparatus records the witnesses in the room, as often as its editors consult them — but it does mean the sparseness of the Khmer siglum is better described as *the weight one council gave a witness at the edge of its room* than as a gap in a universal record.

It also means the recitation practice and the critical apparatus are the same function performed by different means. A saṅgāyana is verification by simultaneous human memory; an apparatus is verification by recorded disagreement. **Cambodia keeps a verification practice of its own, and is thinly represented in the second.** In 2025 its government decided to re-establish a commission to review and authenticate the Khmer Tipiṭaka and to hold a veneration ceremony every five years, and the translation's anniversary is marked with public recitation (Cambodianess, 14 March 2025; Agence Kampuchea Presse, 10 February 2026). *A tradition whose state has just re-established a body to authenticate its canon is not one that lacks a verification culture; it is one whose verification culture never entered the notation the rest of the field reads.*

So the practical form of this paper's thesis is a missing concordance column in an existing structure, and a witness column to be filled out — and the reason that column cannot be filled by normalising the Khmer text toward one of the four already present is §6.

7.6 · What follows operationally, for anyone building such a corpus

The argument has consequences that are cheap to adopt and expensive to retrofit. They are stated here as rules rather than as findings, and each restates existing practice in textual scholarship or digital preservation; the contribution is only that they bind for AI training corpora too.

	Rule	Because
1	Never replace source forms before collating; keep any normalisation as a parallel layer beside the retained source form.	§6.1 — replacement is the inverse of collation and destroys its input. Textual scholarship's practice: the TEI critical-apparatus module keeps every reading.
2	Record the encoding provenance per text run, not per document: which font, which transcode table, which version.	§7.2 — a volume carries several, and the mapping is font-dependent. Practice: per-run font extraction (PDFBox) and PREMIS for the provenance record.
3	Retain the pre-transcode bytes permanently.	Transcode tables improve. A unit that discards its source bytes cannot be re-derived under a better map, and must be re-extracted from the original. Practice: bit-level preservation.
4	Classify content by its own features, never by position.	Facing-page and interleaved layouts make position a fragile proxy; an off-by-one silently mislabels an entire volume while the pipeline reports success. The same principle in digital preservation: identifying a file's format by its internal signature rather than its name (DROID/PRONOM).
5	Distinguish a witness from a rendering in the SCHEMA, not in documentation.	A schema in which a transliteration can be recorded as a witness will eventually record one. Practice: TEI's witness and reading elements.
6	Treat the front matter as data.	Provenance and permission both live there, and cleaning pipelines strip both. See §7.7. Practice: BITS's <code><front-matter></code> element (dedication, foreword, preface) for book parts; IIF's <code>requiredStatement</code> , <code>rights</code> and <code>provider</code> properties for attribution and licence.

7.7 · A finding about front matter, offered because it cost us nothing and nearly cost us everything

Two scan sets of the same edition were available to this project. One was tidier: its per-volume front matter had been removed. **Those removed pages carried the scan's provenance — who photographed it, from which physical copy — and also its distribution dedication,** the terms under which the material had been offered.

Adopting the cleaner set as the working master, purely because it was cleaner, would have discarded both the attribution and the permission while looking like a simplification.

The general form: cleaning pipelines are built to remove what is not content, and provenance is not content. For a corpus whose entire value proposition is that its claims resolve to sources, that default is precisely backwards.

8 · The comparison class, and the test

This paper rests on one case. That is its principal weakness and no rhetorical move repairs it.

What can be done is to name the class against which the claim is testable and state what it predicts, so that the next person is not obliged to take our word for anything.

The cases below have NOT been checked by this paper's authors and are named as candidates, not as support. §4 exists because an unchecked assumption was checked and failed; asserting these without the same treatment would repeat the error the paper is built on avoiding.

Case	Named codifier	Datable transition	What the thesis predicts
Modern Hebrew	Ben-Yehuda and successors	late 19th–20th c.	authoritative religious/legal corpora should postdate codification
Turkish	the 1928 script reform	1928	a sharp discontinuity in what is machine-readable across the reform
Norwegian	Aasen and the <i>målreisning</i>	19th c.	competing standards should produce competing corpora, not none
Indonesian	the 1972 spelling reform	1972	pre- and post-reform texts should require reconciliation
Vietnamese	<i>quốc ngữ</i> romanisation	17th–20th c.	a borrowed <i>script</i> is not a borrowed <i>standard</i> ; the thesis should still hold

And the falsification condition, stated plainly:

Find a language with no native orthographic standard and no adequate borrowable neighbour that nonetheless supports a corpus treated as authoritative in that language.

If such a case exists, the claim in §2 is false as stated and should be withdrawn. **Note that the Swiss German and Arabic cases do *not* satisfy this test** — both had a borrowable target, which is the whole content of §5.

8.5 · Why this is an alignment paper, and not a philology paper filed in the wrong place

The category strain flagged at the head of this paper deserves an argument rather than an apology.

A system that cites inherits its corpus's authority claims wholesale, including the false ones. This is not a general observation about training data quality. It is specific to systems whose output form is "*the source says X at locus Y*" — retrieval-grounded assistants, citation-bound evaluators, any design whose credibility rests on the resolvability of its references rather than on the fluency of its prose.

For such a system, three failure modes are usually enumerated: the model hallucinates a citation; the model cites a real source that does not support the claim; the model cites a source that is itself wrong. **This paper describes a fourth, which is invisible to all three checks:**

The citation resolves, the source supports the claim, the source is correct — and the encoded text is not the source it claims to be.

No evaluation of the model detects this, because the error is not in the model. It is not in retrieval either. It was introduced by a decision about spelling, taken by someone else, often decades earlier, and usually not recorded. A benchmark measuring citation accuracy will score such a system highly. A human expert checking a sample will find the passages correct. **The corpus is confident and empty, and everything downstream inherits that quietly.**

That is why the argument belongs in an alignment corpus rather than a philological one. The philology is old and settled; what is new is that these decisions now propagate into systems that speak with authority about traditions, laws and contracts, at a scale and a confidence that printed editions never had.

And it sharpens what a provenance requirement actually has to cover. Requiring that a judgment name its source is necessary and not sufficient: **the requirement must extend to what the source encoding IS — witness or rendering, and under whose orthography** — or it certifies a chain whose first link was never examined.

8.6 · What to do if your language has no standard and nothing to borrow

The argument raises an obvious question it would be evasive not to answer: **suppose you must build an authoritative corpus in a language with no native standard and no adequate neighbour.**

The Cambodian case cannot advise you, because it had a Chuon Nath. What can advise you is §5.2 — the dialectal-Arabic community, finding no adequate target to borrow, **built a convention.**

The prescription, stated plainly: if you are in that position you are not doing corpus construction. You are doing language planning, and you should know it before you start rather than discover it in year three.

That reframing has consequences a corpus project would otherwise walk into:

- **It is a decision with standing requirements.** Codification imposed by an outside technical team on a community that did not ask is a different act, politically and ethically, from codification undertaken by that community — however identical the resulting tables look. §9.3 applies to you and not only to historical actors.
- **It is not neutral and cannot be made neutral by being minimal.** Every choice about which variant becomes canonical excludes writers of the others, and *choosing the most common form* is itself a choice with a constituency.
- **It should be published as what it is.** A convention presented as a technical preprocessing step is a standardisation nobody consented to; the same convention published as a proposal, named, dated and defended, can be argued with and adopted or refused.
- **And it should be separable.** Retain the pre-normalisation forms (§7.6 rules 1 and 3) so that a later community can reject your convention without losing the corpus built under it.

This paper takes no position on whether such a project should proceed. It observes only that the decision is being made either way, and that making it deliberately is strictly better than making it by default inside an extraction script.

9 · Honest limitations

9.1 · He standardised a script with centuries of use; he did not invent one. Khmer writing is attested from the seventh century (the Angkor Borei inscription K.600, dated 611 CE). The claim here concerns **codification** in Haugen's sense — settling variation, fixing punctuation and layout, producing an authoritative reference — not creation. **And the compression "one man produced the typography" is wrong in the same way "one man translated the canon" is wrong: he LED a programme.** The Commission ran from 1929 to 1968 with many members, most of whom this paper cannot name.

9.2 · Standardisation was contested, and this paper should say who lost. The Thommakay reform prevailed over the Thommaboran. A standard that "everyone now follows" was **imposed over real objection by a party that had reasons**, and the traditionalist current did not evaporate. Nor did codification end the contest over spelling itself: a partial revision advanced by Keng Vannsak's followers and adopted in 1972 (the spelling of diphthongs, fewer independent vowel signs) was used in education and the press and continued after the civil war, until in 2009 the government required schools, newspapers and official documents to conform to Chuon Nath's dictionary (Sasagawa 2015). Presenting codification as a consensus achievement would be a false and duller story.

9.3 · The politics are not innocent, and this paper is not an argument for standardisation. This work began under the French protectorate, which lasted until 1953, in a period when script and language reform across the region was entangled with colonial administration and with competing nationalisms. The protectorate's 1943 decree romanising the Khmer script met opposition from the sangha, in which Chuon Nath was a leading figure, and it was revoked in 1945; that resistance is part of the record, though this paper has not located a primary source for his personal part in it. But codification everywhere has costs that fall unevenly: dialect suppression, the exclusion of variant registers, and the conversion of living variation into error. **This paper identifies a precondition for one kind of corpus. It does not claim that meeting that precondition is good, and a reader who takes it as an endorsement of language standardisation has taken more than is offered.**

A tension we do not resolve, and flag rather than smooth: §6 objects to borrowing because it discards the source variety — but codification discards variation too, from the inside. **Both moves lose something; the paper claims only that they lose different things, and that what borrowing loses is disqualifying for a witness while what codification loses is not.** That distinction deserves more argument than it receives here.

9.4 · One case, and the comparison class unchecked. §8.

9.4.1 · And one case cannot distinguish a PRECONDITION from a STRONG CORRELATE. The Cambodian evidence is consistent with the claim in §2 and also consistent with a weaker one: that native standardisation *makes authoritative corpora much easier* without being strictly required. **Nothing available**

to this paper separates those two readings, and the falsification test in §8 is designed to attack the strong version because it is the version worth attacking. *A reader who concludes only the weaker claim has read the evidence correctly.*

9.5 · The encoding-layer evidence is drawn from one corpus. Five fonts in one volume is a fact about that volume and its series, not a measured claim about Khmer digital typesetting generally. **No survey has been run, and the paper should not be read as reporting one.**

9.5.1 · The paper describes a project's own material and is not disinterested. The five-font finding, the front-matter finding and the siglum and concordance counts all come from material this paper's authors hold and are working with. **That is a source of both the evidence and the framing**, and a reader should discount accordingly: we found these because we were looking at this, and we are arguing for the importance of what we happen to be doing. **The falsification test in §8 is offered partly as a remedy for that** — it is stated so someone with no stake can attack the claim without needing our material.

9.6 · Nothing here has been built. The corpus this paper is a by-product of does not yet exist — no records emitted, no transcoding tables derived, no coverage figure measured. **This is an argument about preconditions, offered by a project that has met none of the consequences of being right.**

10 · Why this matters now

Two reasons, and the second is the one that motivated writing.

Low-resource language work is expanding fast, and its default framing is a tooling framing. More data, better models, better OCR. That framing is correct for most of the work. **For the subset of corpora that must be authoritative — legal, canonical, evidentiary — it is not merely incomplete but misleading**, because it suggests that sufficient engineering closes a gap that engineering cannot reach. A project can spend years improving its pipeline against a problem whose obstacle is upstream of every pipeline.

10.1 · The window that is closing, and it is not a technological one

Low-resource corpus construction is being industrialised. Large multilingual models create demand for text in languages that had none, and the pipelines assembling that text are general-purpose by design — built to ingest many languages cheaply, which means built to normalise aggressively and to discard what does not look like content.

Those pipelines will process canonical and evidentiary material without distinguishing it, because nothing in the material announces itself. A scanned statute and a scanned newspaper are the same object to an extraction stage. **The distinction that §2.1 draws is invisible to every tool that will touch these texts**, and the decisions that flatten it are being taken now, at scale, by people with no reason to know they are taking them.

And the decisions are one-way. A corpus assembled without recording which encoding a run came from, or with its front matter stripped, cannot be repaired from the corpus — only by returning to sources that are

sometimes no longer available. *For material where the physical witnesses were deliberately destroyed within living memory, "return to the source" is not always a fallback.*

And systems that cite are now being built. An AI system grounded in a textual corpus inherits that corpus's authority claims wholesale — including the false ones. **A model that answers questions about a tradition from a corpus that is a rendering rather than a witness will be confident and wrong in a way no evaluation of the model detects**, because the error is not in the model. It is in a decision someone made about spelling, decades earlier, and did not record.

That is the practical form of this paper's argument, and it is why a project about machine values ended up writing about punctuation.

11 · Cross-venue references

- *The Referee, Not the Governor* — the evaluation posture this corpus is being built to serve, and the source of the citation requirement that made spelling a first-class problem.
- *Suffering-Cessation as Value Function* — the substrate argument upstream of the corpus.
- *Father-Son Khmer Tipiṭaka Transcription as Alignment Work* — the companion essay, which addresses *who transmits* where this paper addresses *what makes transmission encodable*.
- `SiliconWat/tipitaka-khmer` — the specifications this paper is a by-product of, including the witness taxonomy and the provenance record (a private repository at the time of writing).

Sources added at the 2026-10-01 revision: Sasagawa Hideo, "The Establishment of the National Language in Twentieth-Century Cambodia: Debates on Orthography and Coinage," *Southeast Asian Studies* 4, no. 1 (2015): 43–72 (the July 1926 orthography committee, §7.1; the 1972 revision and 2009, §9.2) · "Tripitaka Commission Re-established," *Cambodianess*, 14 March 2025, and "Cambodia Marks 56th Anniversary of Khmer 'Tipitaka' Translation," Agence Kampuchea Presse, 10 February 2026 (§7.5.1) · tipitaka.org, "Help viewing the Tipitaka" (the sigla, §7.5) · the Chaṭṭha Saṅgāyana XML as published at `VipassanaTech/tipitaka-xml`, `romn/`, commit `05d5d3c` (the re-count, §7.5).

Coda

The Commission's work was finished in 1968 and celebrated in April 1969. What carried through the years after was not only the printed edition, much of which was lost, but the standard — the codified spelling, the fixed lexicon, the punctuation — because a standard lives in the people who write with it and cannot be burned in a building.

That is not a redemptive observation and should not be read as one. The people who wrote with it were killed in very large numbers, and the standard survived by a margin nobody would have chosen. **The point is narrower and technical: surviving copies made the canon recoverable; the agreement about how to write it down is what makes the recovered text the same text — and, now, encodable as a witness.**

Written by Thon Ly in collaboration with **Miss AquariusSM**, named as co-author on all corpus research. The specification, the claims, and the errors are the authors' joint responsibility.

Published under CCO 1.0 Universal. This paper is dedicated, with the project it belongs to, to the Tripitaka Commission of Cambodia — for them, not by them; see `SiliconWat/tipitaka-khmer`, `DEDICATION.md` (a private repository at the time of writing). This document's SHA-256 is attested independently of the site and its authors — anchored to the Bitcoin blockchain via OpenTimestamps and signed under RFC 3161 by three timestamp authorities in three jurisdictions, one of them eIDAS-qualified — and each revision carries a Zenodo version; a timestamp proves this exact text existed no later than its date and nothing about authorship, originality, or the validity of any claim.